

Dr. Alexander Sherikov

Expert Autonomous Systems Software Engineer

CONTACT INFORMATION

United Arab Emirates

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PERSONAL INFORMATION

Website sherikov.net

Latest CV github.com/asherikov/cv

Other github.com/asherikov

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HIGHLIGHTS

- Fullstack autonomous systems software developer and R&D engineer with a PhD in robot control. Contributed to development of multicopters (up to 40kg), ground vehicles, quadrupedal and full-size humanoid robots. ROS user since 2012.
- Wrote code in C++/Python under Linux for more than 20 years, implemented and enforced automated quality and SITL pipelines to minimize the number of incidents and field trial costs.
- Over 15 years of international experience spanning 6 countries and diverse environments: academic, startup, and corporate. Extensive background in startups: both failed and successful, from inception through product certification, from developer to lead.

EXPERIENCE

2024 – present Senior Robotics Developer, Technology Innovation Institute, UAE

- Integrated and developed UAV onboard software for real-time 3D mapping, and motion planning with obstacle avoidance, deploying on UAV platforms in field trials.
- Contributed to development of drone control stack: reviewed architecture and merge requests, ensured software quality.

2023 – 2024 Autonomy Lead, BaseTrack, UAE

- Documented, maintained, and deployed existing autonomous driving stack for trucks.
- Supported live vehicle demonstrations including sensor integration and software validation.

2022 – 2023 Reinforcement Learning Engineer, Keybotic, Norway (remote contract)

- Trained whole-body locomotion control policies for quadruped robots using parallel GPU-accelerated Isaac-Gym simulations.
- Deployed trained models to test-bench setup using ONNX.

2019 – 2022 Senior Autonomy Engineer, Sevendof, Norway

- Architected and maintained UAV onboard software stack on top of various flight controllers (DJI, PX4) on a range of electric (youtu.be/07FoSaGxQ9g) and hybrid-powered (youtu.be/R63JKgDqbnE) multicopters.
- Established software quality processes including static analysis, automated testing, and CI pipelines, resulting in zero vehicle losses due to software issues over a span of multiple years.
- Designed and maintained software cross-compilation and deployment workflow.
- Implemented 3D volumetric mapping and sensor integration for real-time environment perception and autonomous navigation.
- Managed software team as an acting software lead.

2017 – 2019 Software & Control Engineer, PAL Robotics, Spain

- Developed an in-house rigid body dynamics simulator and implemented identification, motion planning, and inverse kinematics/dynamics controllers for Thalos humanoid robot.
- Supported by “Torres Quevedo Program (PTQ)” grant ciencia.gob.es/stfls/eSede/Ficheros/2018/RESOLUCION_TORRES_QUEVEDO-2017-1.pdf

2016 – 2017 Research Engineer, INRIA, France

- Designed and implemented HUMOTO, a C++ software framework for optimization-based whole-body controllers of humanoid robots bip-team.github.io/humoto/

2012 – 2016 Doctoral Student, INRIA, France

- Developed model predictive controllers for humanoid robot balancing and locomotion.
- Participated in “Horizon H2020 COMANID” research project (cordis.europa.eu/project/id/645097) focused in humanoid robot applications in airplane assembly tasks.

2012 – 2012 Software Developer, Örebro University, Sweden

- Implemented a path-tracking model predictive controller for an autonomous forklift truck.

- Contributed to “SAUNA” project on safe autonomous navigation for industrial vehicles.
oru.se/english/research/research-projects/rp/?rdb=p693

2009 – 2010 System & Network Administrator, InfoLan LLC, Russia

- Administered FreeBSD servers and configured networking hardware for an Internet service provider serving several thousand clients.

EDUCATION

2012 – 2016 PhD, Automatic Control and Production Systems

Thesis Balance preservation and task prioritization in whole body motion control of humanoid robots
github.com/asherikov/phd-thesis/raw/master/asherikov-phd-thesis.pdf

School University of Grenoble, France

Courses 2015, Sliding-Mode Control: Theory and Applications, Aussois, France
2014, Numerical Optimal Control, Freiburg, Germany

2010 – 2012 MS, Robotics and Intelligent Systems

Thesis Model predictive control of a walking bipedal robot using online optimization
github.com/asherikov/ms-thesis/raw/master/asherikov-ms-thesis.pdf

School Örebro University, Sweden

2003 – 2008 Engineer (MS), Information Systems and Technologies

Thesis Application of multidimensional data structures for indexing of NetFlow records

School Petrozavodsk State University, Russia

PERSONAL OPEN-SOURCE PROJECTS

SOFTWARE ARCHITECTURE, DESIGN, CONVENTIONS

- github.com/asherikov/codeincomplete: Coding conventions tailored for robotics and autonomous systems
- github.com/asherikov/hiearch [python]: Diagram generator with hierarchical decomposition and views

DEVELOPMENT TOOLS

- github.com/asherikov/ccws [sh, make]: ROS development and CI framework for cross-compilation, testing, linting, documentation, and binary package deployment
- github.com/asherikov/shoggoth [docker]: Self-hosted agentic development multitool intended for personal use

MATH AND CONTROL

- github.com/asherikov/qpmad [C++]: High-performance quadratic programming solver based on Goldfarb-Idnani algorithm
- sherikov.net/Projects/naowalk.html [C++]: Walking controller for Nao humanoid robot

UTILITY LIBRARIES

- github.com/asherikov/ariles [C++]: Reflection/serialization library supporting ROS parameters, YAML, JSON, XML, and more
- github.com/asherikov/thread_supervisor [C++]: C++17 thread supervisor with automatic restart of failed or finished threads
- github.com/asherikov/intrometry [C++]: Telemetry collection library for real-time data monitoring, similar to `pal_statistics` and `data_tamer`.

SKILLS

SOFTWARE DEVELOPMENT

C++

Standards and libraries: C++11/14/17/20, STL, Boost, POSIX

Parallel computations: multithreading (pthreads), GPU (CUDA/Thrust)

Compilation: clang, gcc, nvcc, cross-compilation, emulated compilation

Debugging: gdb, lldb, strace

Performance optimization: callgrind, gprof

Testing: googletest, googlemock, Boost UTF, ctest

Other languages

sh/bash: various automation scripts

python: general scripting, reinforcement learning, numerical computations, data processing

Octave/MATLAB: numerical computations, scientific computing

Source code, build, packaging

Version control systems: git, SVN

Web-based SCM: GitHub, GitLab, Gitea, BitBucket

Build optimization: ccache

Build automation: catkin, colcon, cmake, make, autotools

Packaging: FreeBSD ports, debian/apt, fpm, CloudSmith, conan, vcpkg, pypi

Agentic (AI-assisted) coding

Environments: Qwen-Code, OpenCode

Concepts: MCP, skills, memory

Model management: Ollama (self-hosted and cloud)

Software quality

Continuous integration: GitLab, GitHub/Gitea Actions, Jenkins, Travis

Static analysis: cppcheck, clang-tidy, scan-build, pylint, flake8

Dynamic analysis: gcc/clang sanitizers (address, undefined, thread), valgrind

Documentation: doxygen, Confluence, markdown, pandoc, L^AT_EX, GitLab/GitHub/Gitea pages

Process

Architecture: system, interface, and protocol design, UML

Source code: maintenance and evolution of legacy code, dependency migration

Methodologies: Agile, Scrum, Kanban, Git workflows (branching, merge requests, reviews)

Task management: Jira, Trello, Redmine

Leadership: mentorship, planning, estimation, problem solving, project management

UNIX SYSTEMS AND NETWORKING

Operating systems: FreeBSD, Ubuntu, Alpine Linux

Containerization/emulation: docker, docker-compose, qemu, systemd-nspawn, VirtualBox, proot, chroot

Service management: systemd, dinit, OpenRC

Computer networks: TCP/IP, VLAN, routing, switching

Network services: DHCP (dnsmasq), DNS (unbound), SMTP, VPN (OpenVPN), ssh, web (nginx)

Network utilities: ip/ifconfig, ss/netstat, wireshark

Time synchronization: NTP, PTP

CLI: xargs, sed, grep, screen/tmux, jq, vim, ...

ROBOTICS AND AUTONOMOUS SYSTEMS

UAV

Flight controllers: PX4, DJI, ArduPilot

Messaging: mavlink, UAVCAN, OpenCyphal, mavros, Micro-XRCE-DDS-Agent

Aircraft tracking: ADS-B

Ground control (GCS): QGroundControl

Simulation

Simulators: Microsoft AirSim, Gazebo, IsaacGym

Visualization: RViz, OpenSceneGraph

Validation: CI automated, headless Software-In-The-Loop (SITL) testing

Volumetric mapping

Sensors: lidars (Ouster, Livox), altimeters, radars

Data formats: point clouds (PCL), range images

Mapping: OpenVDB, OctoMap, Bonxai

Navigation

Sensors: GPS (ublox, SwiftNav Piksi), INS (VectorNav)

High-precision positioning: RTK, NTRIP, base-rover setup

Geodetic libraries: geopandas, pyproj, shapely, GeographicLib, OSMnx

Other

Frameworks: ROS, ROS2, Nao SDK

Motion planning: OMPL

Behavior: behavior trees, state machines
Model representation: URDF, SDF
Messaging: protobuf, CAN, mqtt
Telemetry: time-series databases, MCAP, Grafana, PlotJuggler
Hardware platforms: Raspberry Pi, NVIDIA Jetson, Pixhawk

MATH AND CONTROL

Numerical optimization: quadratic programming (qpOASES, QuadProg++, ipopt, qpnm) sequential quadratic programming, prioritized least squares (LexLS), linear complementarity problems (siconos)
Modeling and control: rigid body modeling of robots (RBDL), basics of friction and collision, inverse kinematics and dynamics, model predictive control
Numerical libraries: Eigen, numpy, pandas, scipy
Reinforcement learning: PyTorch, optuna, tensorboard, IsaacGym, ONNX
Numerical environments / CAS: Octave/MATLAB, Maxima, ipython
Visualization: matplotlib, graphviz, asymptote

OTHER

Languages: Russian (native), English (fluent)
Driving licence: category LMV (UAE), B (Russia).

ACADEMIC ACTIVITIES

- Reviewer for IEEE T-RO, ICRA, IROS, Humanoids.
- Google Scholar page: scholar.google.fr/citations?user=yV0vGdOAAAAJ&hl=en (over 600 citations).

PUBLICATIONS

- [1] D. J. Agravante, A. Cherubini, A. Sherikov, P.-B. Wieber, and A. Kheddar. “Human-Humanoid Collaborative Carrying”. In: *IEEE Transactions on Robotics* 35.4 (2019), pp. 833–846. DOI: [10.1109/TR0.2019.2914350](https://doi.org/10.1109/TR0.2019.2914350). URL: <https://hal-lirmm.ccsd.cnrs.fr/lirmm-01311154>.
- [2] D. J. Agravante, A. Sherikov, P.-B. Wieber, A. Cherubini, and A. Kheddar. “Walking pattern generators designed for physical collaboration”. In: *IEEE ICRA*. 2016.
- [3] N. Bohórquez, A. Sherikov, D. Dimitrov, and P.-B. Wieber. “Safe navigation strategies for a biped robot walking in a crowd”. In: *IEEE-RAS International Conference on Humanoid Robots*. 2016.
- [4] S. A. Homsy, A. Sherikov, D. Dimitrov, and P.-B. Wieber. “A hierarchical approach to minimum-time control of industrial robots”. In: *IEEE ICRA*. 2016.
- [5] D. Serra, C. Brasseur, A. Sherikov, D. Dimitrov, and P.-B. Wieber. “A Newton method with always feasible iterates for Nonlinear Model Predictive Control of walking in a multi-contact situation”. In: *IEEE-RAS International Conference on Humanoid Robots*. 2016.
- [6] H. Andreasson, A. Bouguerra, M. Cirillo, D. Dimitrov, D. Driankov, L. Karlsson, A. Lilienthal, F. Pecora, J. Saarinen, A. Sherikov, and T. Stoyanov. “Autonomous Transport Vehicles: Where We Are and What Is Missing”. In: *Robotics Automation Magazine, IEEE* 22.1 (2015).
- [7] C. Brasseur, A. Sherikov, C. Collette, D. Dimitrov, and P.-B. Wieber. “A robust linear MPC approach to online generation of 3D biped walking motion”. In: *IEEE-RAS International Conference on Humanoid Robots*. 2015.
- [8] D. Dimitrov, A. Sherikov, and P.-B. Wieber. “Efficient resolution of potentially conflicting linear constraints in robotics”. Preprint. 2015. URL: <https://hal.inria.fr/hal-01183003>.
- [9] A. Sherikov, D. Dimitrov, and P.-B. Wieber. “Balancing a humanoid robot with a prioritized contact force distribution”. In: *IEEE-RAS International Conference on Humanoid Robots*. 2015.
- [10] A. Sherikov, D. Dimitrov, and P.-B. Wieber. “Whole body motion controller with long-term balance constraints”. In: *IEEE-RAS International Conference on Humanoid Robots*. 2014.
- [11] D. Dimitrov, A. Sherikov, and P.-B. Wieber. “A sparse model predictive control formulation for walking motion generation”. In: *IEEE/RSJ IROS*. 2011.
- [12] A. Sherikov and Y. Bogoyavlenskii. “The use of multidimensional index structures for NetFlow record processing”. In: *AMICT '07, Proceedings of the Annual International Workshop on Advances in Methods of Information and Communication Technology*. 2007.